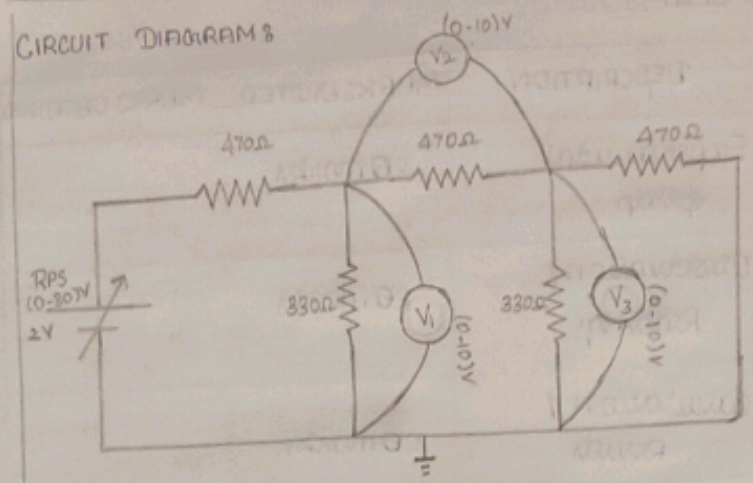
CIRCUIT DIAGRAM:



Exp no: 03	DETERMINATION OF VOLTAGES IN CIRCUIT USING NODAL ANALYSIS
Date: 06/02/2026	

AIM:

To determine the voltages in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED:

Sl.NO	APPARATUS	SPECIFICATIONS	QUANTITY
1	Regulated Power supply	(0-30V)	1
2	Multimeter	---	1
3	Resistor	470Ω, 330Ω	3,2
4	Bread board	---	1

THEORY:

In electric circuits analysis, nodal analysis, node-voltage analysis, or the branch current method is a method of determining the voltage between "nodes" in an electrical circuit in terms of the branch currents.

PROCEDURE:

1) Give connections as per the circuit diagram.

TABULAR COLDUMN :

PARAMETES	THEDRETICAL	PRACTICAL
V_1	0.63	0.66
V_2	0.18	0.19
V_3	0.45	0.47

~~Calcul~~

CALCULATION :

Nodal analysis :

At Node A :

By KCL :

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$\frac{33V_A - 66 + 47V_A + 33V_A - 33V_B}{15570} = 0$$

$$113V_A - 33V_B = 66 \rightarrow (1)$$

At node B

By KCL :

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33V_B - 33V_A + 47V_B + 33V_B = 0$$

ii) Switch ON the power supply, vary the RPS and set a particular input voltage.

iii) Note down the readings of the ammeter and voltmeters and tabulate them.

iv) Vary the RPS for different input voltages and note down the readings of all the meters.

v) Reduce the RPS to its minimum value and switch OFF the supply.

vi) Using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

$$113V_B - 33V_A = 0$$

$$113V_B = 33V_A$$

$$V_A = \frac{113V_B}{33} \rightarrow (2)$$

Substitute (2) in (1)

$$113\left(\frac{113V_B}{33}\right) - 33V_B = 66$$

$$\frac{12769}{33}V_B - \frac{1089}{33}V_B = 66$$

$$V_B = \frac{2178}{11680}$$

$$V_B = 0.186V$$

$$V_A = \frac{113}{33}V_B = \frac{113}{33} \times 0.18$$

$$V_A = 0.638V$$

$$V_1 = V_A = 0.63V$$

$$V_2 = V_B = 0.18V$$

$$V_3 = V_A - V_B = 0.63 - 0.18$$

$$= 0.45V$$

MARKS SPLIT-UP:

SI.NO	DESCRIPTION	MARKS ALOTTED	MARKS OBTAINED
1.	Experimental setup	20 marks	20
2.	Observation / Readings	20 marks	20
3.	Calculation / result	20 marks	20
4.	Viva	20 marks	18
5.	Record	20 marks	18
6.	Total	100 marks	~~20~~ 96

RESULT:

Thus, the nodal analysis is verified practically and theoretically. The resultant voltages for 2V supply are

a) The voltage V_1 is 0.63V

b) The voltage V_2 is 0.18V

c) The voltage V_3 is 0.45V